

Specification & Documentation

Release Date: 30 June 2026 | **Source Repository:**

<https://github.com/pgvector/pgvector/tree/v0.8.4>

1. Architecture & Capabilities

pgvector is an open-source vector similarity search extension for PostgreSQL. Release v0.8.4 provides production-grade support for HNSW (Hierarchical Navigable Small World) and IVFFlat index structures, L2 distance, cosine distance, inner product, and L1 distance operators.

2. Key Features Evaluated in Benchmark

- **Partial Indexing (WHERE Clause Integration):** Enables predicate pushdown by creating specialized indices on filtered sub-relations: `CREATE INDEX ON items USING hnsw (embedding vector_cosine_ops) WHERE location_id = ...;`
- **Iterative Index Scan:** Optimizes post-filtering by dynamically relaxing search depth (`hnsw.ef_search`) to overcome filtered-ANN recall collapse.
- **Cross-Engine Benchmark:** Evaluated in Table 8 & Table 9 against Milvus and Weaviate under identical hardware, selectivity, and dimensional constraints.